Regression in Practice

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Regression in Practice

This R day is going to have you apply different regression strategies presented in topic 04. An emphasis will be placed on interpreting and presenting regression results. For this assignment, I have assembled a dataset consisting of homes in Massachusetts.

There are a lot of variables in this dataset, so please see the separate codebook for details.

Because this data is large (almost 2 million rows), I am using the `parquet` file format. You can load this with the `read_parquet` function from the `arrow` package. You might have to change your working directory in quarto to the folder where this `.qmd` file is located.

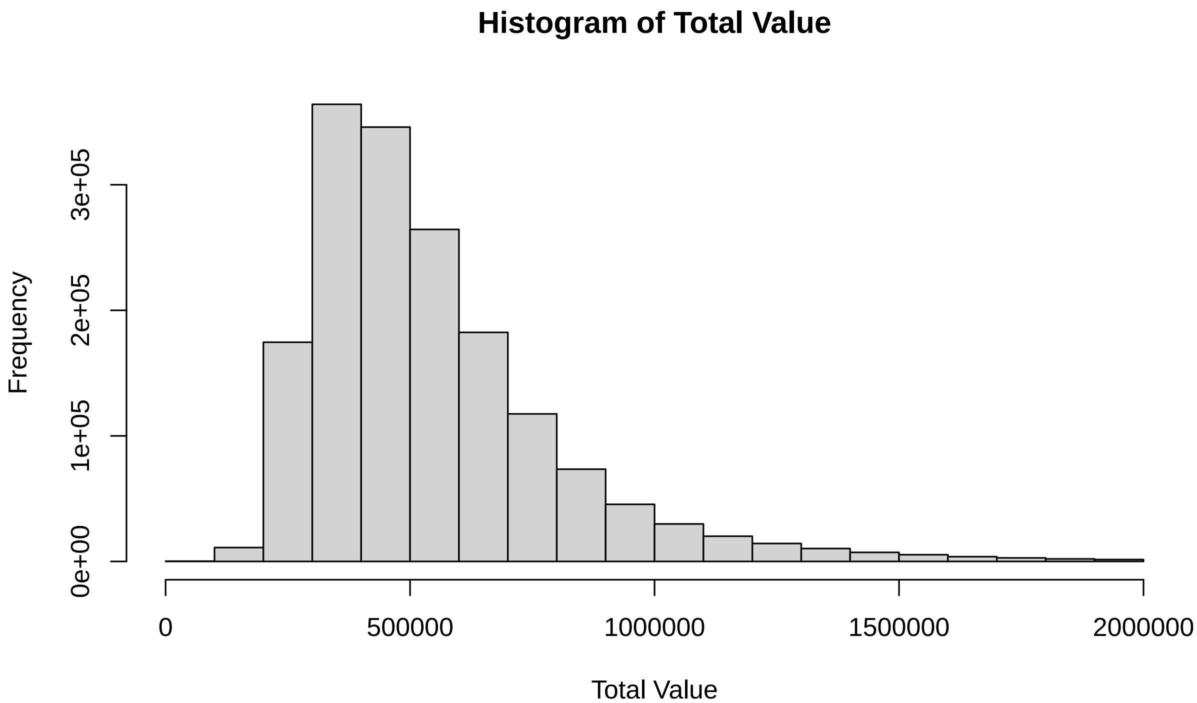
```r
library(ggplot2) # install.packages("ggplot2")
library(fixest) # install.packages("fixest")
library(arrow) # install.packages("arrow")

df <- read_parquet("data/MA_parcels_sample.parquet")
```

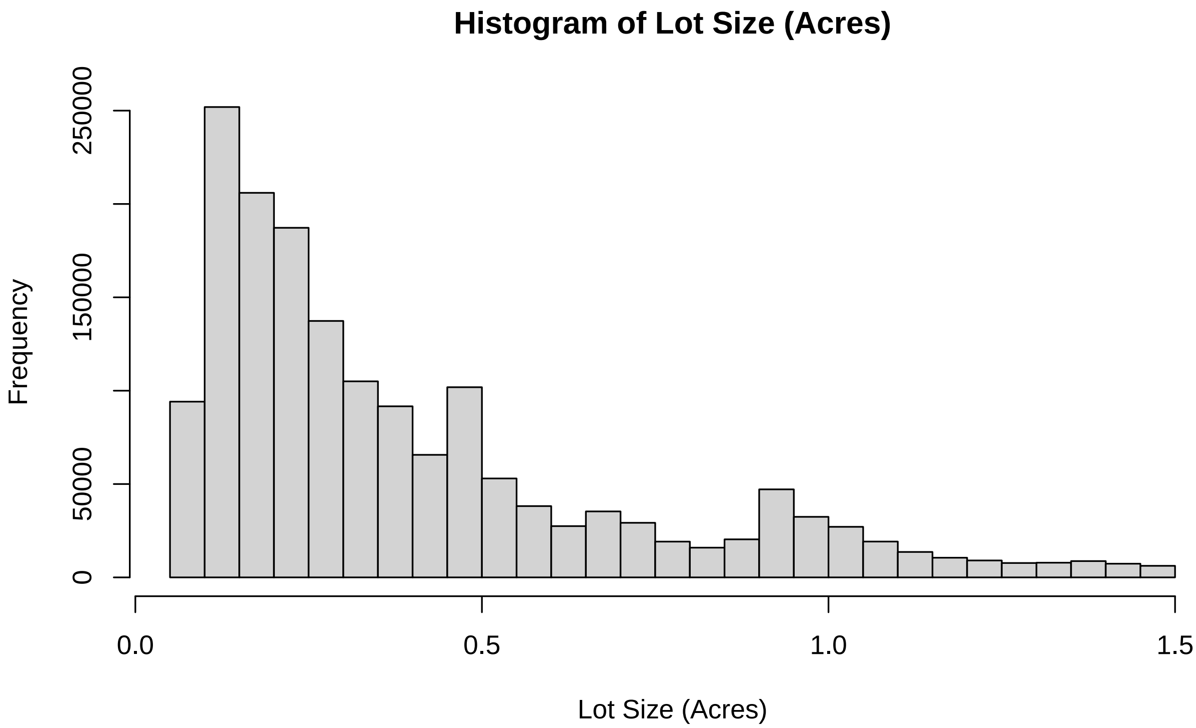
Basic exploration of variables

One of the main variables we will use in this analysis is the assessed value of the home (`total_value`) and the size of the land (`lot_size_acres`). Like good data analysts, let's create histograms of both variables. Comment on the distribution.

```r
# Basic histograms of total_value and lot_size_acres
hist(df$total_value, main = "Histogram of Total Value", xlab = "Total Value")
```




```
hist(
  df$lot_size_acres,
  main = "Histogram of Lot Size (Acres)",
  xlab = "Lot Size (Acres)"
)
```



Answer:

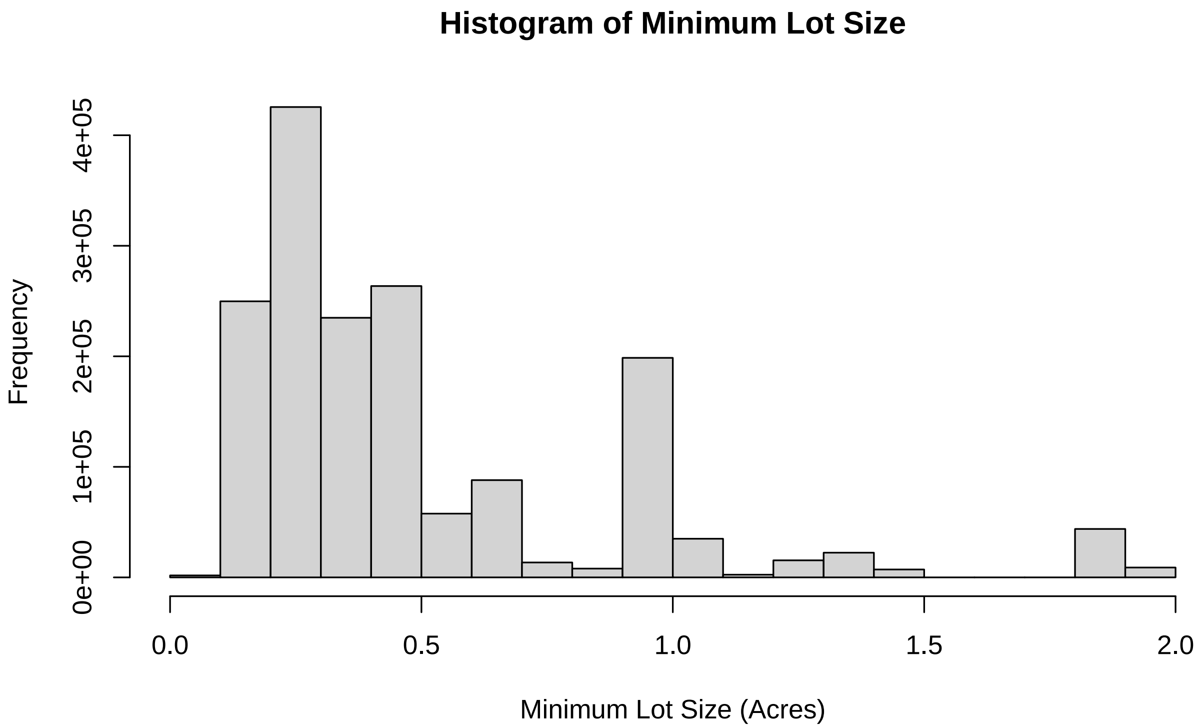
Now, let's look at a few explanatory variables. Use the `table` function to explore the distribution of `n_rooms` and make a histogram of `z_mls_acres`.

```
# Explore n_rooms distribution and histogram of z_mls_acres
table(df$n_rooms)
```

```

     3      4      5      6      7      8      9     10 
  3950  41010 197091 444000 372259 344610 165880 107509 
```

```
hist(
  df$z_mls_acres,
  main = "Histogram of Minimum Lot Size",
  xlab = "Minimum Lot Size (Acres)"
)
```



Answer:

Relationship between lot size and building value

We are going to explore the relationship between the lot size and the value of the building. We will make a scatter plot to start our analysis. However, there is way too many points to plot in the dataset, so we will sample a random 10000 rows.

```
# Sample 25000 rows for analysis
sample_df <- df[sample(1:nrow(df), 25000), ]
```

Bivariate regression

First, regress building value on the lot size linearly using `feols` from the `fixest` package. Interpret the regression coefficient and comment on it's statistical significance. Make sure to use `vcov = "HC1"` to get robust standard errors.

```
# Linear regression of building value on lot size
model1 <- feols(total_value ~ lot_size_acres, data = sample_df, vcov = "HC1")
summary(model1)
```

```
OLS estimation, Dep. Var.: total_value
Observations: 25,000
Standard-errors: Heteroskedasticity-robust
                Estimate Std. Error  t value  Pr(>|t|)
(Intercept)     481295.7    2460.38 195.6181 < 2.2e-16 ***
lot_size_acres  149379.0    5057.05  29.5388 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 251,779.5   Adj. R2: 0.03351
```

Answer:

Forecasting

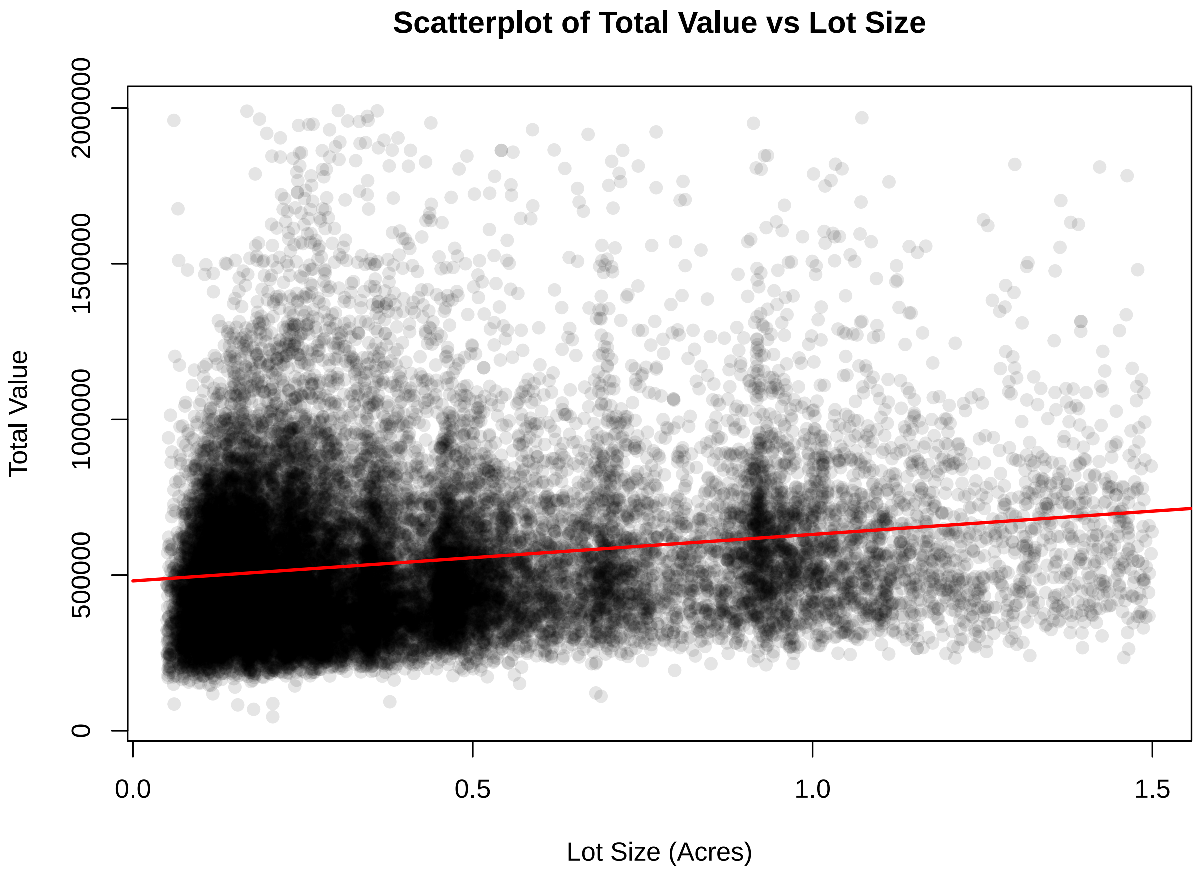
Now, let's try and show what our forecasted values will be. To do so, let's create a `data.frame` containing a grid of values of `lot_size_acres`. We can do that with `seq(0, 4, 0.01)`. Call this `pred_grid`. Create forecasts from your simple linear model using `predict(..., newdata = pred_grid)`. Store the forecasts back into the `pred_grid` data frame.

To create a plot, first copy your `plot` call above to create the sample scatterplot. Then, use the `lines` function to overlay your forecasted line of best fit. Give this line a distinct color and make the line thicker by adding the arguments `col = "red", lwd = 2`.

```r
# Create prediction grid
pred_grid <- data.frame(
  lot_size_acres = seq(0, 4, 0.01)
)

# Generate predictions
pred_grid$predicted_value <- predict(model1, newdata = pred_grid)

# Plot scatter with regression line
plot(
  sample_df$lot_size_acres,
  sample_df$total_value,
  main = "Scatterplot of Total Value vs Lot Size",
  xlab = "Lot Size (Acres)",
  ylab = "Total Value",
  pch = 19,
  col = rgb(0, 0, 0, 0.1)
)
lines(pred_grid$lot_size_acres, pred_grid$predicted_value, col = "red", lwd = 2)
```



Answer:

Forecasting with confidence intervals

Now, use your regression to predict the assessed value for a home on a 1.5 acre lot. Add the option `se.fit = TRUE` to your `predict` call to get a standard error for the predicted value. Present your result with a 95% confidence interval.

```
# Predict for 1.5 acre lot with standard error
pred_1.5 <- predict(
  model1,
  newdata = data.frame(lot_size_acres = 1.5),
  se.fit = TRUE
)
pred_1.5$fit
```

```
[1] 705364.3
```

```r
pred_1.5$se.fit
```

```
[1] 5921.96
```

```r
# Calculate 95% CI
ci_lower <- pred_1.5$fit - 1.96 * pred_1.5$se.fit
ci_upper <- pred_1.5$fit + 1.96 * pred_1.5$se.fit
ci_lower
```

```
[1] 693757.2
```

```r
ci_upper
```

```
[1] 716971.3
```

Answer:

Polynomial

Estimate a new model with a 4th-order polynomial of `lot_size_acres`. You can do this simply with the `poly` function. Forecast this model using `pred_grid` and create a new forecast plot similar to above. Add lines for both the linear fit (in red) and the polynomial fit (in blue)

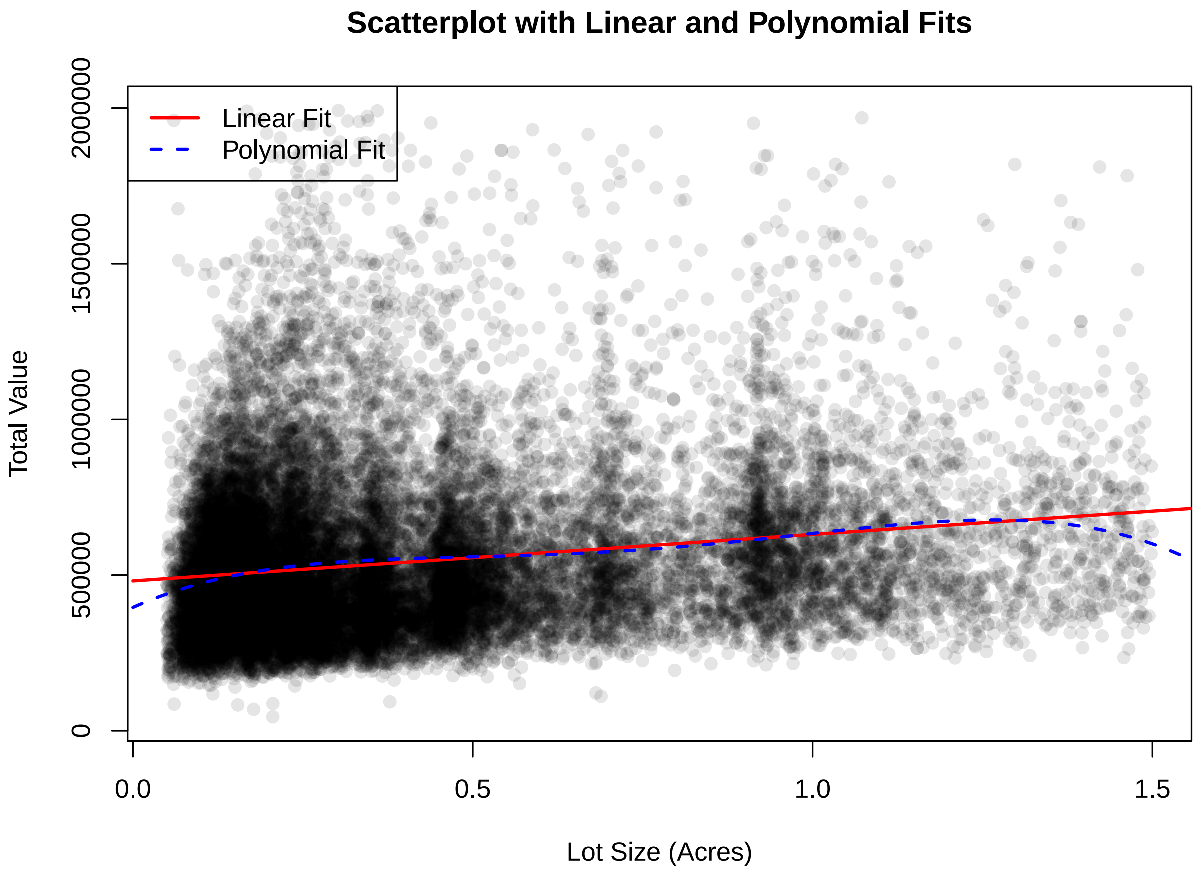
```r
# 4th-order polynomial regression
model_poly <- feols(
  total_value ~ poly(lot_size_acres, 4),
  data = sample_df,
  vcov = "HC1"
)
summary(model_poly)
```

```
OLS estimation, Dep. Var.: total_value
Observations: 25,000
Standard-errors: Heteroskedasticity-robust
                           Estimate Std. Error   t value   Pr(>|t|)
(Intercept)                540750.9    1589.97 340.10067  < 2.2e-16 ***
poly(lot_size_acres, 4)1  7417163.6  250956.94  29.55552  < 2.2e-16 ***
poly(lot_size_acres, 4)2 -1228609.3  247931.28  -4.95544 7.2637e-07 ***
```

```
poly(lot_size_acres, 4)3   663581.1  253128.93   2.62151 8.7593e-03 **
poly(lot_size_acres, 4)4 -1783126.8  253839.27  -7.02463 2.2013e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 251,371.7   Adj. R2: 0.036522
```

```r
# Generate polynomial predictions
pred_grid$predicted_poly <- predict(model_poly, newdata = pred_grid)

# Plot both linear and polynomial fits
plot(
  sample_df$lot_size_acres,
  sample_df$total_value,
  main = "Scatterplot with Linear and Polynomial Fits",
  xlab = "Lot Size (Acres)",
  ylab = "Total Value",
  pch = 19,
  col = rgb(0, 0, 0, 0.1)
)
lines(
  pred_grid$lot_size_acres,
  pred_grid$predicted_value,
  col = "red",
  lwd = 2,
  lty = 1
)
lines(
  pred_grid$lot_size_acres,
  pred_grid$predicted_poly,
  col = "blue",
  lwd = 2,
  lty = 2
)
legend(
  "topleft",
  legend = c("Linear Fit", "Polynomial Fit"),
  col = c("red", "blue"),
  lty = c(1, 2),
  lwd = 2
)
```



Answer:

Adding other predictors

Now, let's try to add the number of rooms and the building area to our polynomial model. Let's create indicatro variables for each value of `n_rooms` using the `i` function. Control for building_area linearly.

```
# Model with indicator variables for n_rooms and linear building_area
model_final <- feols(
  total_value ~ poly(lot_size_acres, 4) + i(n_rooms) + building_area,
  data = sample_df,
  vcov = "HC1"
)
summary(model_final)
```

```
OLS estimation, Dep. Var.: total_value
```

```
Observations: 25,000
Standard-errors: Heteroskedasticity-robust
                              Estimate   Std. Error   t value   Pr(>|t|)
(Intercept)                252202.3297  28951.36194  8.711242  < 2.2e-16 ***
poly(lot_size_acres, 4)1  -411538.7121 221415.44137 -1.858672 6.3085e-02 .
poly(lot_size_acres, 4)2     2277.9867 198327.84380  0.011486 9.9084e-01
poly(lot_size_acres, 4)3   390691.8519 202132.72551  1.932848 5.3266e-02 .
poly(lot_size_acres, 4)4 -1715527.1577 205707.89486 -8.339627  < 2.2e-16 ***
n_rooms::4                 -27265.4628  29360.32573 -0.928650 3.5308e-01
n_rooms::5                 -11939.7921  28964.69501 -0.412219 6.8018e-01
n_rooms::6                  19571.0129  28937.33550  0.676324 4.9884e-01
n_rooms::7                  65940.5194  29032.88506  2.271236 2.3141e-02 *
n_rooms::8                 102219.0131  29172.14567  3.503994 4.5914e-04 ***
n_rooms::9                 172809.0085  29583.81570  5.841336 5.2425e-09 ***
n_rooms::10                177598.6361  29993.72495  5.921193 3.2379e-09 ***
building_area                  89.1097      1.62963 54.681026  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 203,835.7   Adj. R2: 0.366264
```

Answer the following questions:

1. What is the omitted category for `n_rooms`?
2. Interpret in words the coefficient on `n_rooms::7` and comment on it's statistical significance.
3. We could have run a regression controlling linearly for the number of rooms (treating it as a continuous measure). Do your estimates for `i(n_rooms)` make you comfortable with making that simplifying assumption?
4. What is the estimated return on square footage for homes in Massachusetts? To help put this result in context, multiply this coefficient by the standard deviation of `building_area`. Your interpretation will be a for "1 standard deviation increase in `building_area`"

Answer:

Fixed Effects

One concern we might have with interpreting our regression results is that the number of rooms a house has is likely related to what town the home is in. Let's add a set of town indicators to limit our regression to compare homes "in the same town", so to speak. It turns out there are 98 different towns in our sample, so this will end up slowing down our regression quite a bit. Instead, we will specify the `town` variable after a vertical bar, `|`, in our formula. It will look like `y ~ x | town`. `feols` will estimate these effects very quickly, but note their coefficients are not displayed in the summary.

Modify the previous regression model to add town "fixed effects" (i.e. a set of indicators for each town).

```
# Model with town fixed effects
model_fe <- feols(
  total_value ~ poly(lot_size_acres, 4) + i(n_rooms) + building_area | town,
  data = sample_df,
  vcov = "HC1"
)
summary(model_fe)
```

```
OLS estimation, Dep. Var.: total_value
Observations: 25,000
Fixed-effects: town: 97
Standard-errors: Heteroskedasticity-robust
                              Estimate   Std. Error    t value   Pr(>|t|)
poly(lot_size_acres, 4)1  5967460.7274 225961.94475  26.409140  < 2.2e-16 ***
poly(lot_size_acres, 4)2 -3077894.8931 174477.35839 -17.640655  < 2.2e-16 ***
poly(lot_size_acres, 4)3  1287171.1330 161111.79637   7.989304 1.4154e-15 ***
poly(lot_size_acres, 4)4  -972641.8111 154750.85111  -6.285211 3.3284e-10 ***
n_rooms::4                 -44880.1603  27204.39592  -1.649739 9.9009e-02 .
n_rooms::5                 -43179.2782  26933.50707  -1.603181 1.0891e-01
n_rooms::6                 -42038.7760  26932.39371  -1.560900 1.1856e-01
n_rooms::7                 -32003.8639  26973.69267  -1.186484 2.3544e-01
n_rooms::8                  -4795.3052  27043.34719  -0.177319 8.5926e-01
n_rooms::9                  28621.9272  27273.85264   1.049427 2.9399e-01
n_rooms::10                 26059.9766  27563.19115   0.945463 3.4443e-01
building_area                  97.6898      1.68812  57.868863  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 143,895.0     Adj. R2: 0.682962
                  Within R2: 0.427259
```

How does this result change our estimate on `building_area`?

Answer:

Zoning Regulation

Zoning regulations limit the kind of development that can occur on a parcel. We have a bunch of zoning measures including: - `z_mls_acres`, the minimum lot size (the required amount of land needed to build a house). - `z_permit_2fam`, whether 2 family homes are allowed to be built "By Right", "By Permit", or "Not Allowed". - `z_residential`/`z_nonresidential`/`z_mixed` that says what kind of development can be built on a parcel: residential, commercial, or both.

Using regression methods, try to present a case that more stringent regulations (higher minimum lot size) is associated with larger observed lot sizes. You should write up your results in a paragraph and include a figure. Make sure to interpret the statistical significance of your results.

```r
# Regression of lot size on zoning minimum lot size
model_zoning <- feols(
  lot_size_acres ~ z_mls_acres,
  data = sample_df,
  vcov = "HC1"
)
summary(model_zoning)
```

```
OLS estimation, Dep. Var.: lot_size_acres
Observations: 25,000
Standard-errors: Heteroskedasticity-robust
            Estimate Std. Error t value  Pr(>|t|)
(Intercept) 0.142593   0.002634 54.1391 < 2.2e-16 ***
z_mls_acres 0.514295   0.006445 79.7937 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.24474   Adj. R2: 0.392607
```

```r
# Scatterplot of lot size vs zoning minimum
plot(
  sample_df$z_mls_acres,
  sample_df$lot_size_acres,
  main = "Scatterplot of Lot Size vs Minimum Lot Size Requirement",
  xlab = "Minimum Lot Size Requirement (Acres)",
  ylab = "Actual Lot Size (Acres)",
  pch = 19,
  col = rgb(0, 0, 0, 0.1)
)
abline(model_zoning, col = "red", lwd = 2)
```


Scatterplot of Lot Size vs Minimum Lot Size Requirement
Actual Lot Size (Acres)
0.0
0.5
1.0
1.5
0.0
0.5
1.0
1.5
2.0
Minimum Lot Size Requirement (Acres)

Answer: